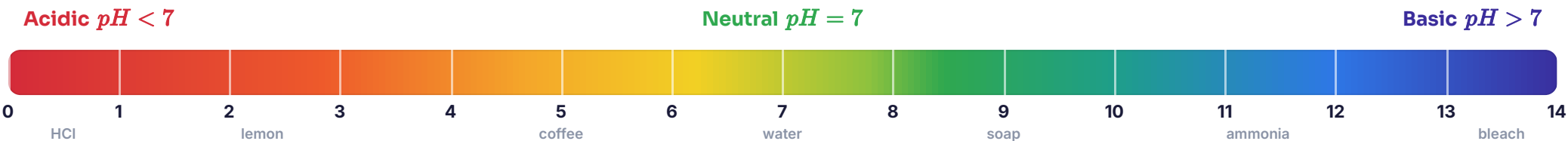


pH The pH Scale at 25°C · universal-indicator colours



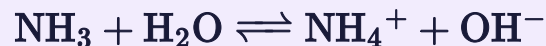
A Arrhenius

- Acid $\rightarrow$ gives H^+ (H_3O^+) in water
- Base $\rightarrow$ gives OH^- in water



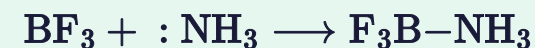
B Brønsted-Lowry

- Acid = proton (H^+) donor
- Base = proton acceptor



C Lewis

- Acid = e^- pair acceptor
- Base = e^- pair donor



± Conjugate Acid-Base Pairs

acid $\leftrightarrow$ conjugate base



base $\leftrightarrow$ conjugate acid

KEY Stronger acid $\Rightarrow$ weaker conjugate base. Differ by a single H^+ .

K_w Ionic Product of Water

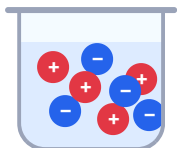
$$K_w = [H^+][OH^-] = 10^{-14}$$

$$[H^+] = [OH^-] = 10^{-7} M$$

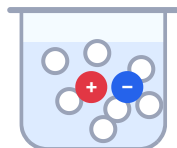
T_↑ Ionisation is endothermic: $T \uparrow \Rightarrow K_w \uparrow$. Neutral $pH < 7$ at high T (still neutral).

$$pH = -\log[H^+] \quad pH + pOH = 14 \quad pK_a + pK_b = pK_w = 14 \quad K_a \cdot K_b = K_w$$

% Strong vs Weak



Strong · $\alpha \approx 1$



Weak · $\alpha \ll 1$

● $\text{HCl}, \text{HNO}_3, \text{H}_2\text{SO}_4, \text{HClO}_4$

● $\text{CH}_3\text{COOH}, \text{HF}, \text{HCN}$

o Ostwald's Dilution Law weak acid, conc C

$$K_a = \frac{C\alpha^2}{1 - \alpha}$$

$$\alpha \ll 1 \Rightarrow$$

$$\alpha = \sqrt{\frac{K_a}{C}}$$

$$[\text{H}^+] = C\alpha = \sqrt{K_a C}$$

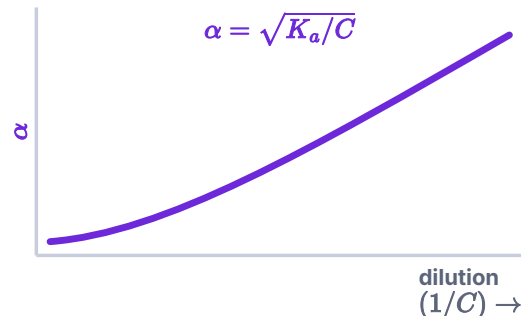
$$\alpha \propto \frac{1}{\sqrt{C}} \quad (\text{dilute} \Rightarrow \alpha \uparrow)$$

f Master pH Formulas

SOLUTION	$[\text{H}^+]$ OR $[\text{OH}^-]$	pH
Strong acid (C)	$[\text{H}^+] = C$	$pH = -\log C$
Strong base (C)	$[\text{OH}^-] = C$	$pH = 14 + \log C$
Weak acid	$[\text{H}^+] = \sqrt{K_a C}$	$pH = \frac{1}{2}(pK_a - \log C)$
Weak base	$[\text{OH}^-] = \sqrt{K_b C}$	$pOH = \frac{1}{2}(pK_b - \log C)$

Weak base $\rightarrow$ then $pH = 14 - pOH$ · approximations valid for $\alpha \ll 1$

α Dilution vs Degree of Dissociation



~ Conjugate & Very Dilute Acid

$$K_a \cdot K_b = K_w$$

$$[\text{H}^+] = \frac{C + \sqrt{C^2 + 4K_w}}{2}$$

n Polyprotic & Levelling

- $K_{a1} \gg K_{a2} \gg K_{a3}$ — pH set by K_{a1} (e.g. H_3PO_4)
- **Levelling:** all strong acids $\rightarrow$ leveled to H_3O^+ in water

? Salt Hydrolysis — Nature of the Salt

	Strong base	Weak base
Strong acid	Neutral · $pH = 7$ NaCl, KNO ₃ — no hydrolysis	Acidic · $pH < 7$ NH ₄ Cl — cationic hydrolysis
Weak acid	Basic · $pH > 7$ CH ₃ COONa — anionic hydrolysis	Depends on K_a vs K_b CH ₃ COONH ₄ — both hydrolyse

WA·SB Basic



$$K_h = \frac{K_w}{K_a}$$

$$h = \sqrt{\frac{K_w}{K_a C}}$$

$$pH = 7 + \frac{1}{2}pK_a + \frac{1}{2}\log C$$

SA·WB Acidic



$$K_h = \frac{K_w}{K_b}$$

$$h = \sqrt{\frac{K_w}{K_b C}}$$

$$pH = 7 - \frac{1}{2}pK_b - \frac{1}{2}\log C$$

WA·WB Both

h is independent of C

$$K_h = \frac{K_w}{K_a K_b}$$

$$h = \sqrt{\frac{K_w}{K_a K_b}}$$

$$pH = 7 + \frac{1}{2}(pK_a - pK_b)$$

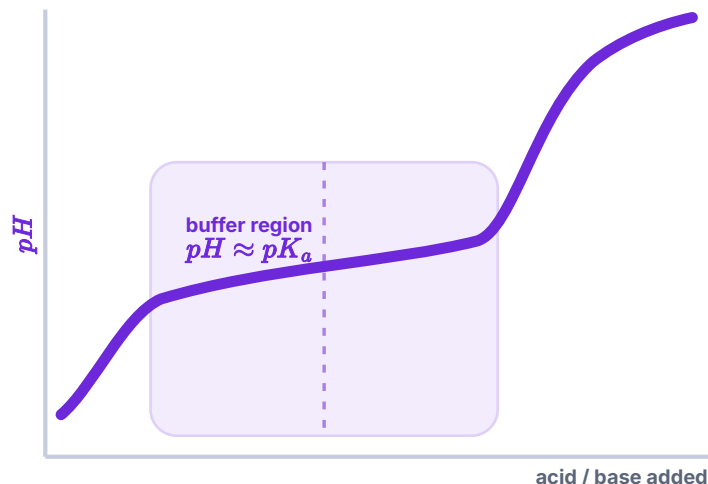
i Read-off

- $K_a > K_b \Rightarrow$ acidic
- $K_a < K_b \Rightarrow$ basic
- $K_a = K_b \Rightarrow$ neutral

TIP Dilution & larger $K_h \Rightarrow$ more hydrolysis h .

B What is a Buffer?

Resists pH change on adding small acid / base or on dilution.



Ac Acidic Buffer weak acid + its salt

HENDERSON-HASSELBALCH

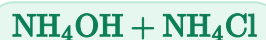
$$pH = pK_a + \log \frac{[\text{salt}]}{[\text{acid}]}$$



$$[\text{salt}] = [\text{acid}] \Rightarrow pH = pK_a$$

Bs Basic Buffer

$$pOH = pK_b + \log \frac{[\text{salt}]}{[\text{base}]}$$



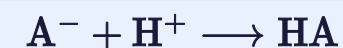
$$pH = 14 - pOH$$

β Buffer Capacity

$$\beta = \frac{\Delta(\text{acid/base})}{\Delta pH}$$

- Max when $[\text{salt}] = [\text{acid}]$
- Useful range $pH = pK_a \pm 1$

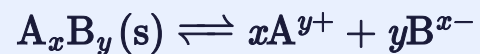
⇌ Buffer Action



BODY Blood $\text{H}_2\text{CO}_3/\text{HCO}_3^-$ holds $pH \approx 7.4$.

Ks Solubility Product

Sparingly soluble A_xB_y :



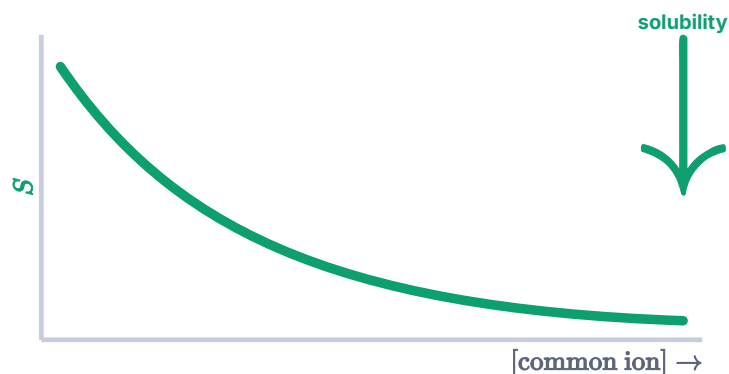
$$K_{sp} = [A^{y+}]^x [B^{x-}]^y$$

$$K_{sp} = x^x y^y S^{(x+y)}$$

S Solubility $S \leftrightarrow K_{sp}$

TYPE	EXAMPLE	K_{sp}	S
AB	$AgCl, BaSO_4$	S^2	$\sqrt{K_{sp}}$
AB_2/A_2B	CaF_2, Ag_2CrO_4	$4S^3$	$\sqrt[3]{K_{sp}/4}$
AB_3/A_3B	$Fe(OH)_3, AlF_3$	$27S^4$	$\sqrt[4]{K_{sp}/27}$
A_3B_2	$Ca_3(PO_4)_2$	$108S^5$	$\sqrt[5]{K_{sp}/108}$

C Common Ion Effect



$$S_{AgCl} = \frac{K_{sp}}{C} \text{ in } C M \text{ NaCl}$$

Q Precipitation — Ionic Product Q vs K_{sp}

$Q < K_{sp}$
Unsaturated
no precipitate

$Q = K_{sp}$
Saturated
equilibrium

$Q > K_{sp}$
Supersaturated
precipitate forms

SELECT Fractional ppt: ion needing lower [reagent] (smaller K_{sp}) precipitates **first**.

In Acid-Base Indicators

Methyl orange	3.1–4.4	
Methyl red	4.2–6.3	
Litmus	5.5–7.5	
Bromothymol blue	6.0–7.6	
Phenolphthalein	8.3–10.0	

pK Indicator Equation

Weak acid HIn : colour of $\text{HIn} \neq \text{In}^-$

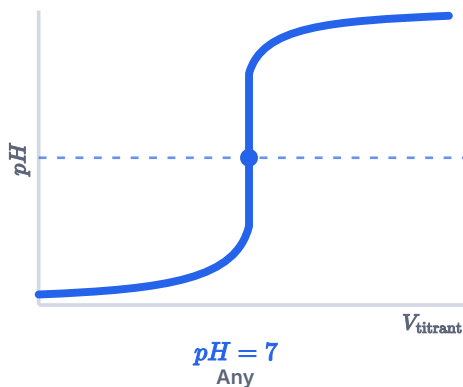


$$\text{pH} = \text{pK}_{\text{In}} + \log \frac{[\text{In}^-]}{[\text{HIn}]}$$

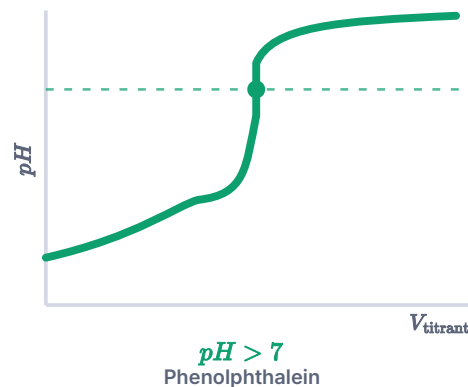
RANGE Colour changes over $\text{pK}_{\text{In}} \pm 1$. Half-equivalence of WA-SB: $\text{pH} = \text{pK}_a$.

T Titration Curves & Indicator Choice

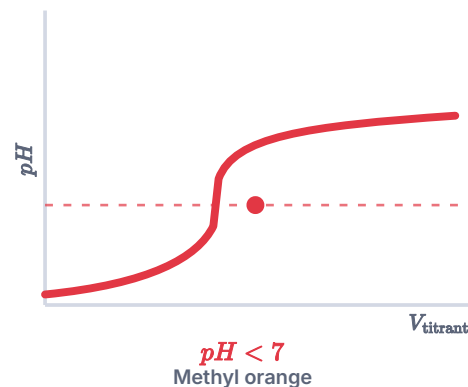
SA – SB



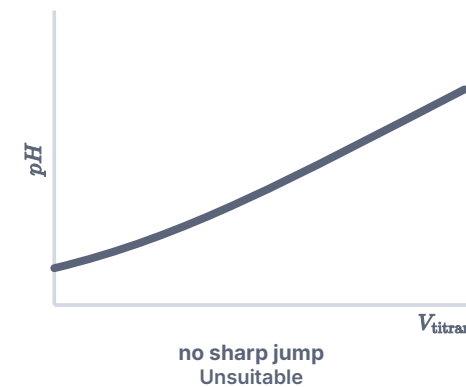
WA – SB



SA – WB



WA – WB



★ Golden Rules — Last-Minute Recall

WATER

$$\text{pH} + \text{pOH} = 14$$

CONJUGATE

$$K_a K_b = K_w$$

WEAK ACID

$$[\text{H}^+] = \sqrt{K_a C}$$

BUFFER

$$\text{pH} = \text{pK}_a + \log \frac{[s]}{[a]}$$

PRECIPITATE

$$Q > K_{sp}$$